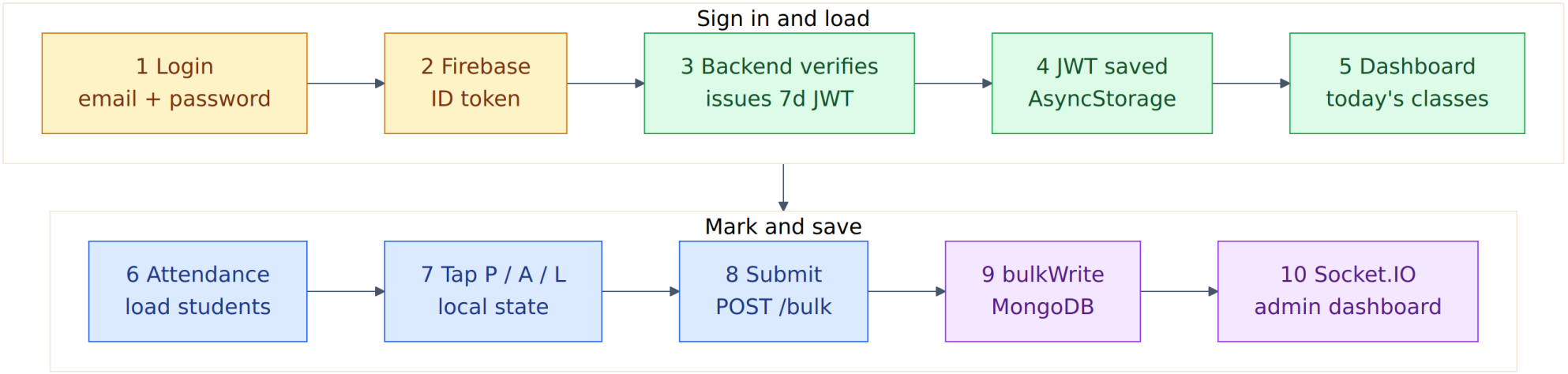
System Design

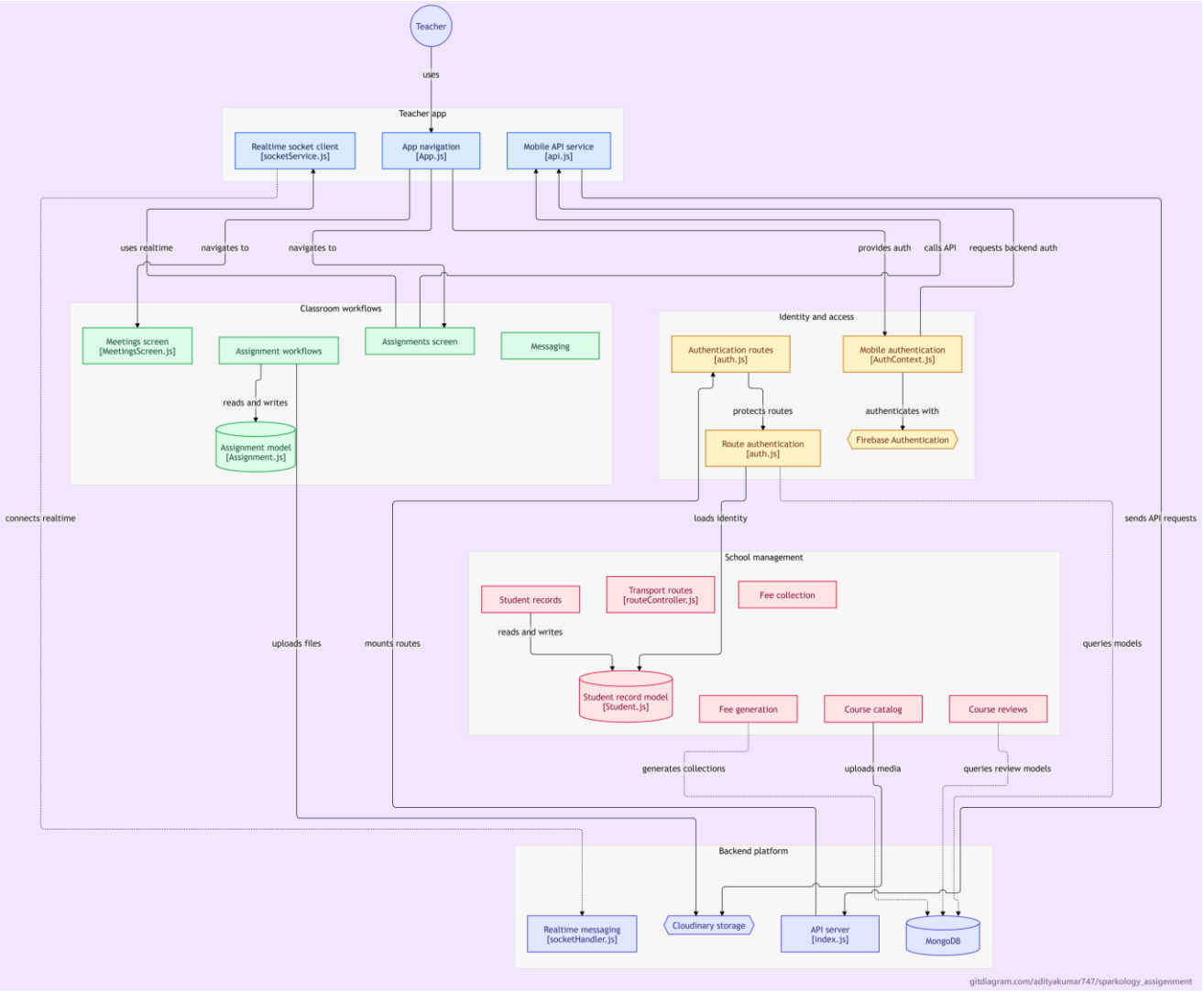
1. End-to-end flow

How one teacher session moves through the system, from sign-in to a live update on the admin dashboard.



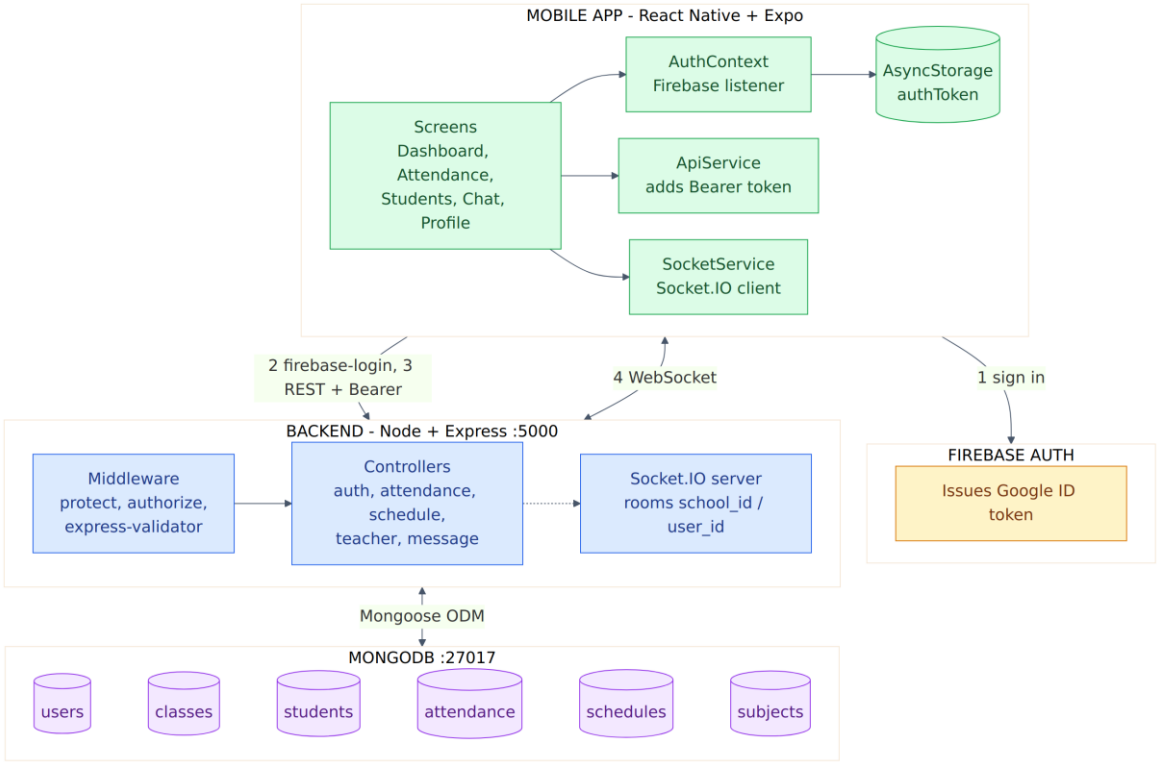
Identity	Transport	Guard	Persistence	Real-time
Firebase proves who you are; backend swaps it for its own 7-day JWT.	ApiService adds the Bearer token to every REST call automatically.	protect() then authorize(): token, active user, role.	Mongoose reads/writes; bulkWrite saves a whole class at once.	Socket.IO pushes attendance_updated to the school room.

2. Repository map



Teacher app	Classroom workflows	Identity and access	School management	Backend platform

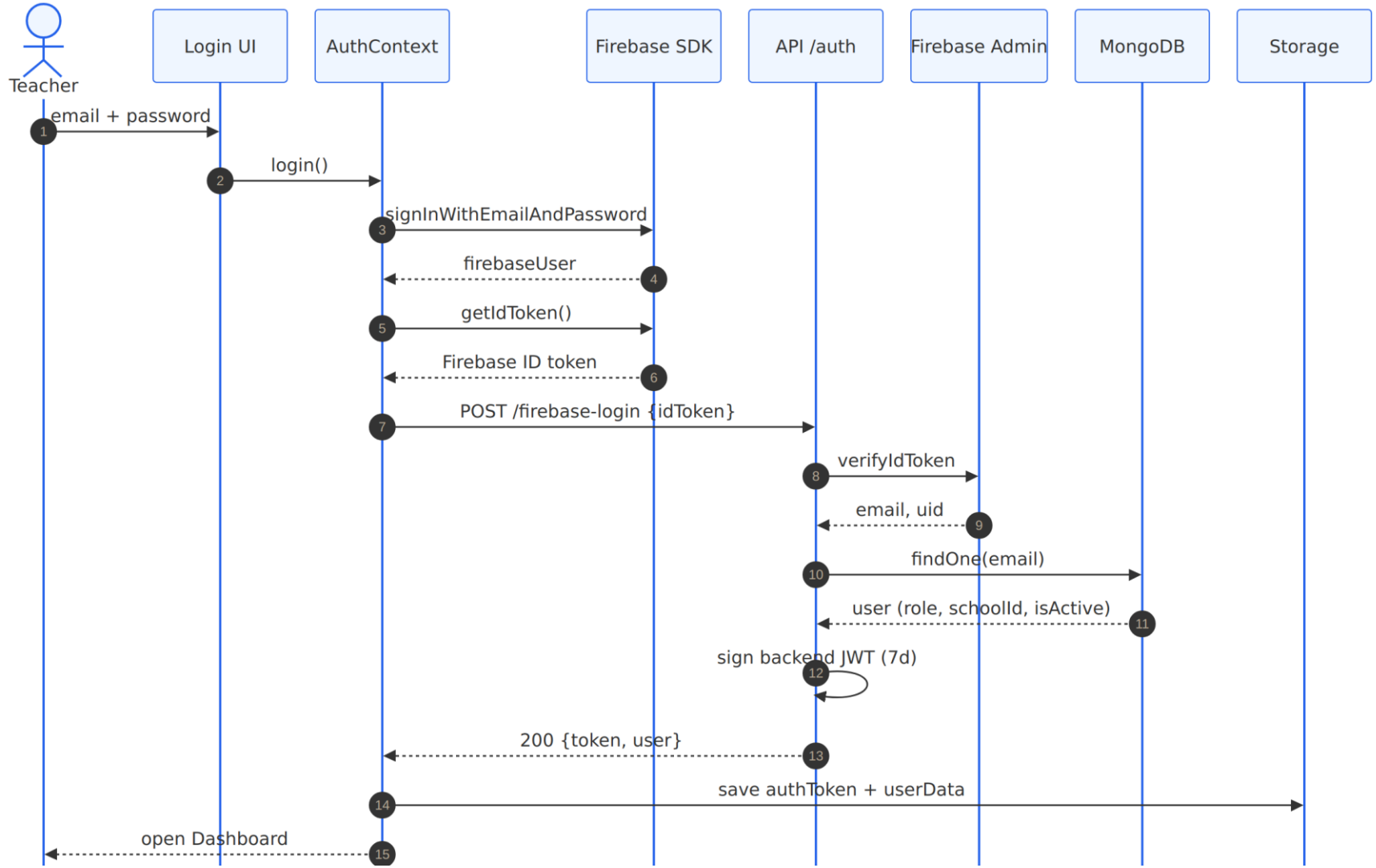
3. System architecture



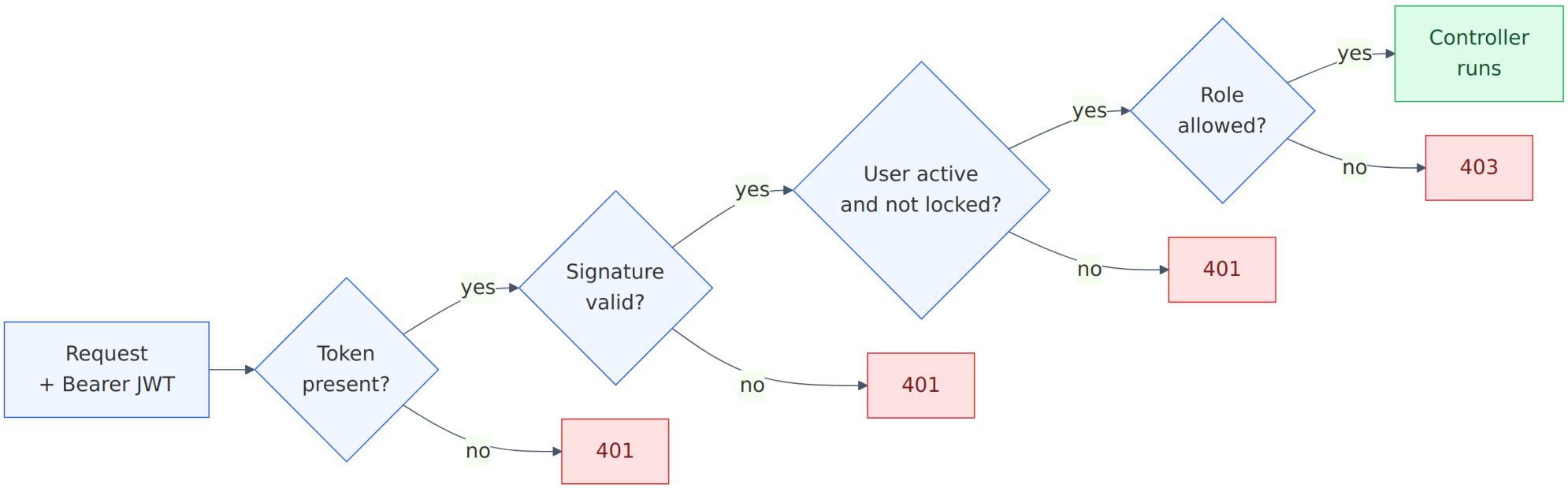
Tier	Technology	Key responsibilities
Mobile client	React Native 0.81, Expo 54, Socket.IO client 4.8	Screens, AuthContext (Firebase session), ApiService (Bearer header), SocketService (live events)
Backend	Node.js, Express 4.18, Socket.IO, jsonwebtoken, express-validator	protect() verifies JWT + isActive + not locked; authorize() enforces role; controllers hold business logic
Database	MongoDB 6/7, Mongoose 8	users, classes, students, attendance, teacherattendances, schedules, subjects
External	Firebase Auth, Cloudinary	Identity (ID token), media uploads

4. Data flow pipelines

4.1 Authentication: POST /api/auth/firebase-login



4.2 What every protected request goes through

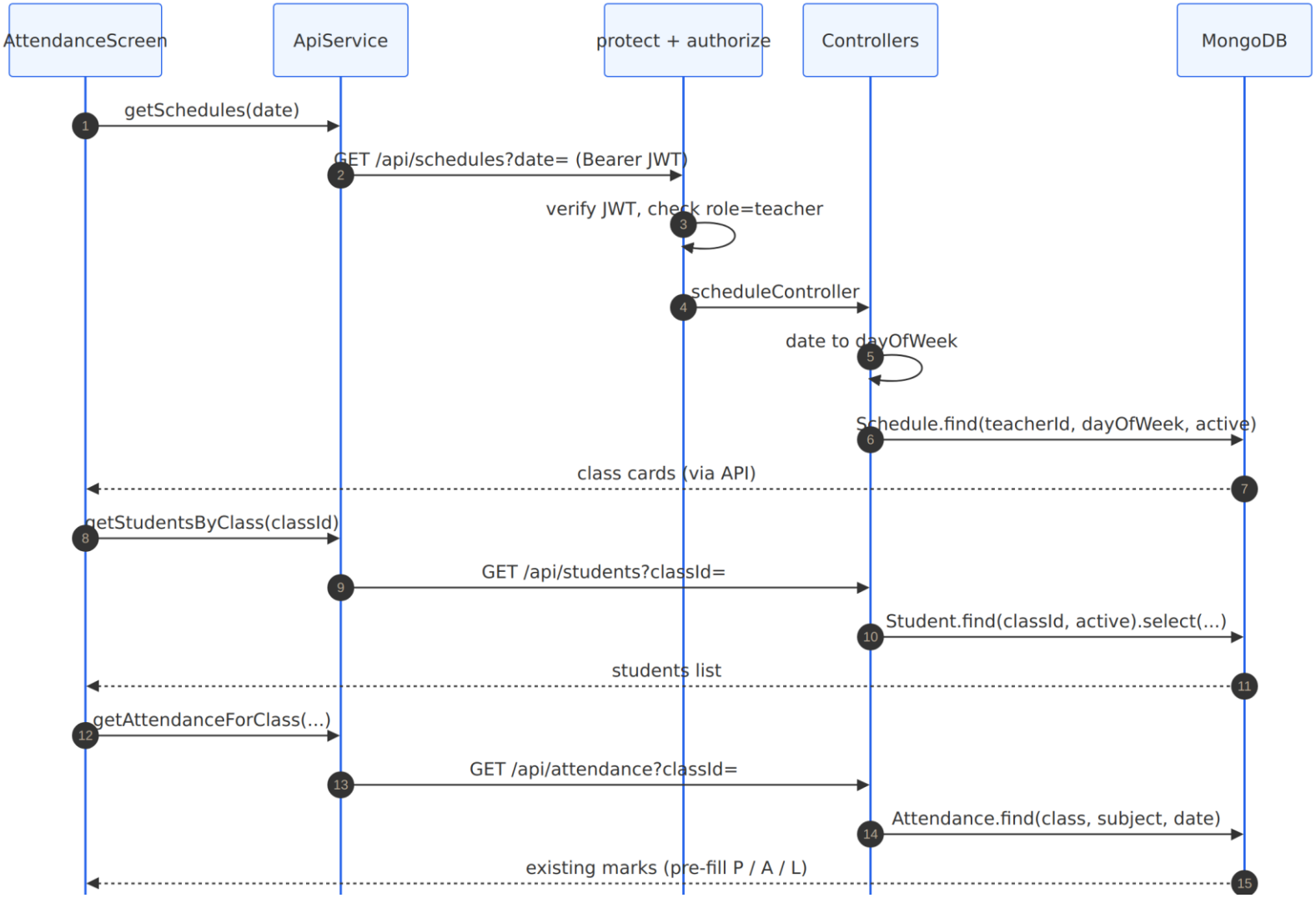


401 Unauthorized
Missing header, bad signature, expired token, deactivated user, or locked account (5 failed logins).

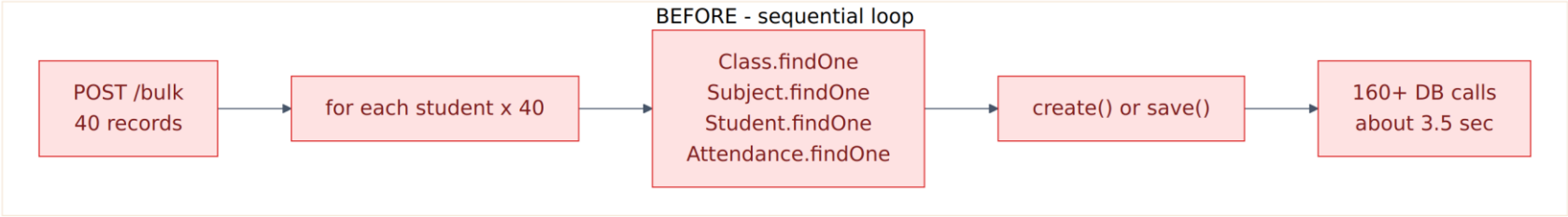
403 Forbidden
Valid user, wrong role - for example a teacher calling an admin-only route.

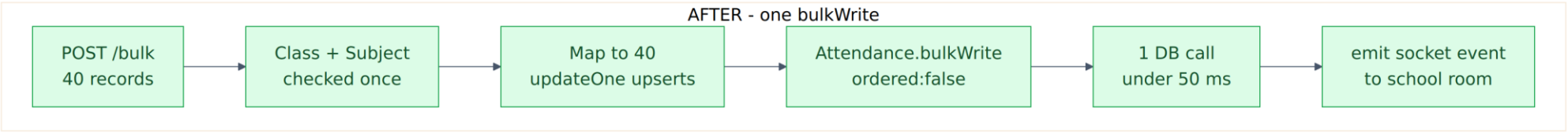
Why old tokens still fail
The user is re-read from MongoDB on every request, so deactivating a teacher takes effect immediately.

4.3 Attendance screen load: schedules, students, existing marks



4.4 Bulk attendance save: before vs after





Why the old loop hurts

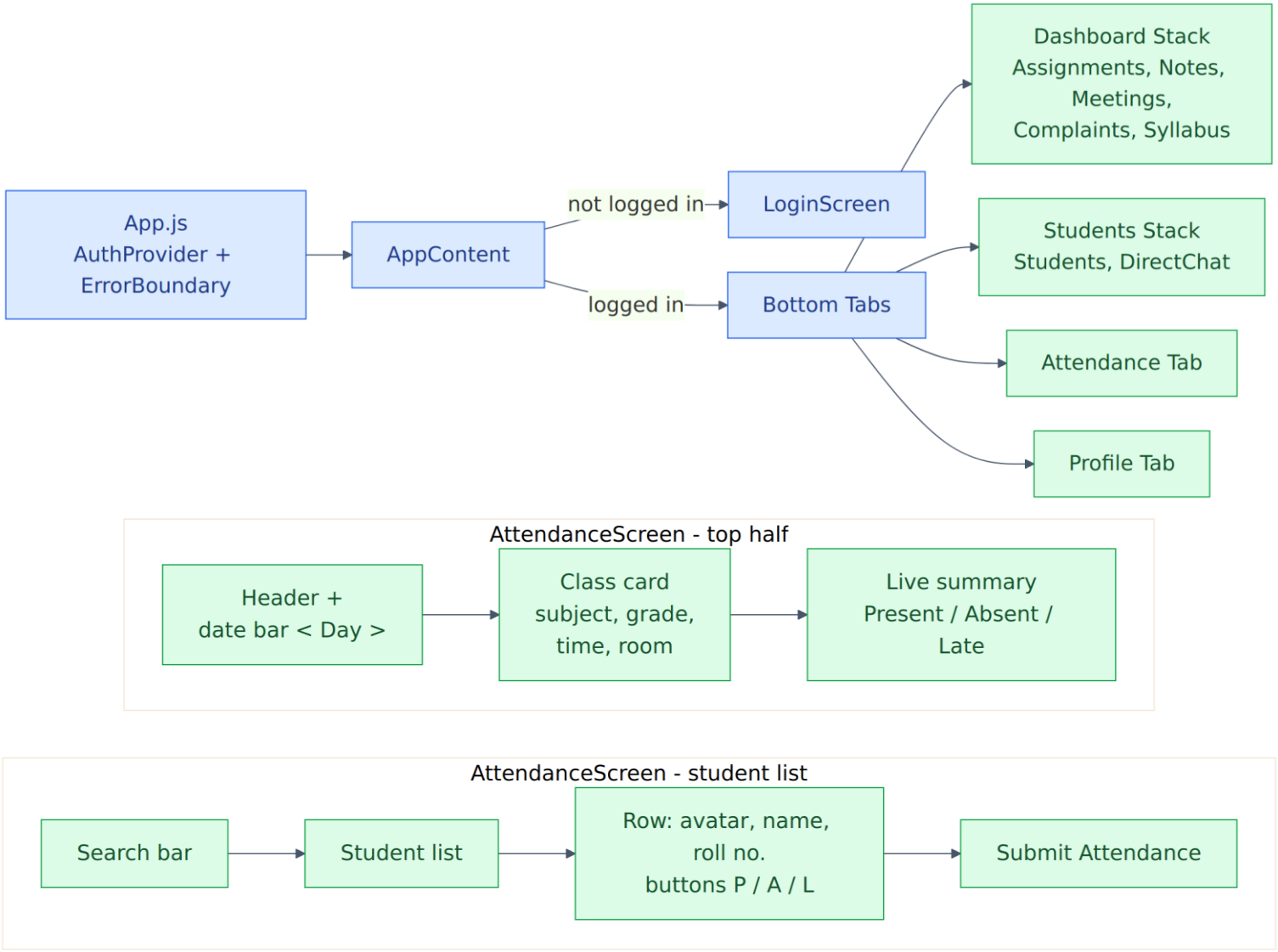
Each student costs 4 queries + 1 write, one after another. 40 students = 160+ round-trips (3-5 s). If the connection drops midway, the class is left half-marked.

Why bulkWrite fixes it

Class and subject are checked once. updateOne + upsert inserts new marks and edits existing ones in a single round-trip, so re-submitting never raises E11000.

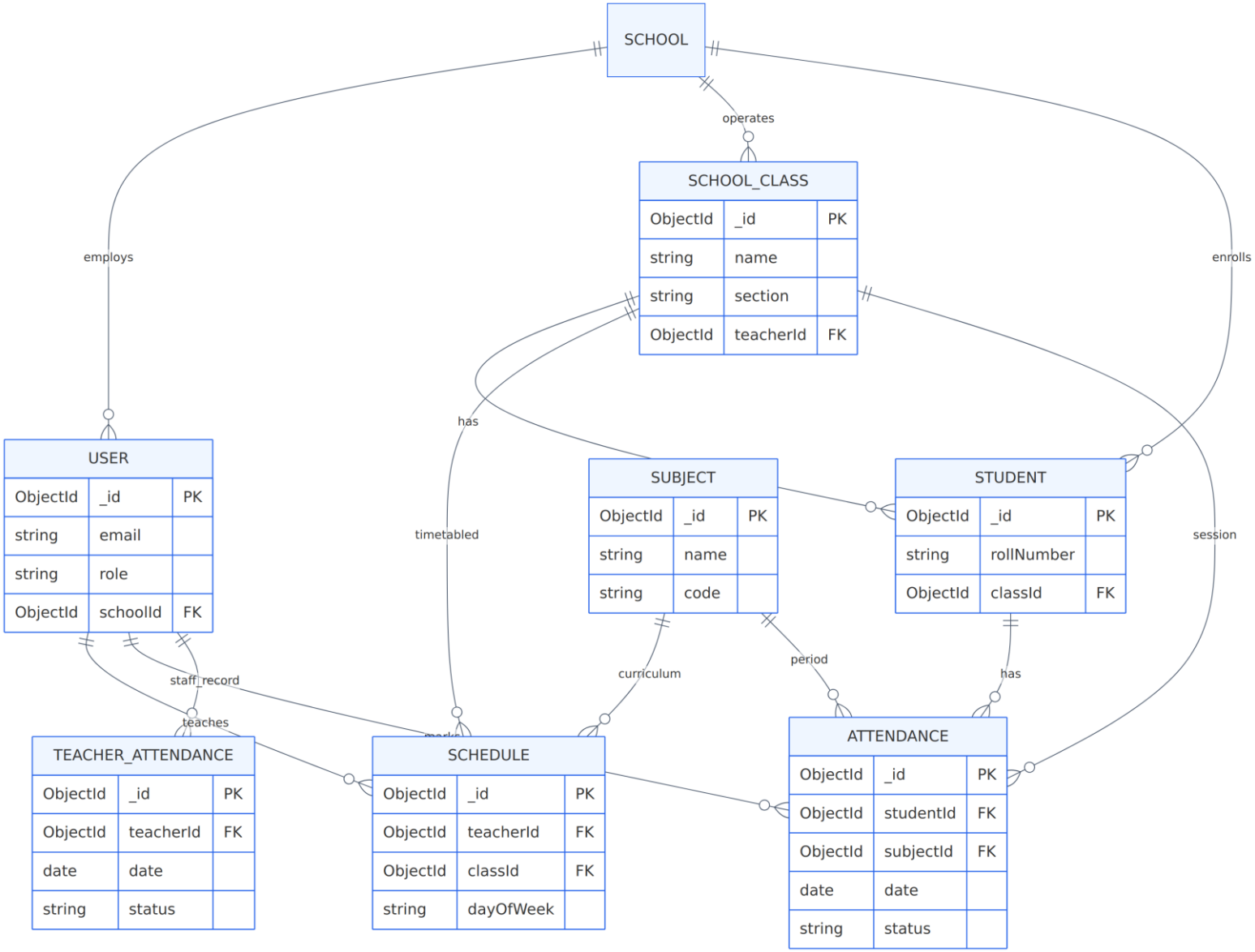
5. App structure and database

5.1 Mobile component hierarchy

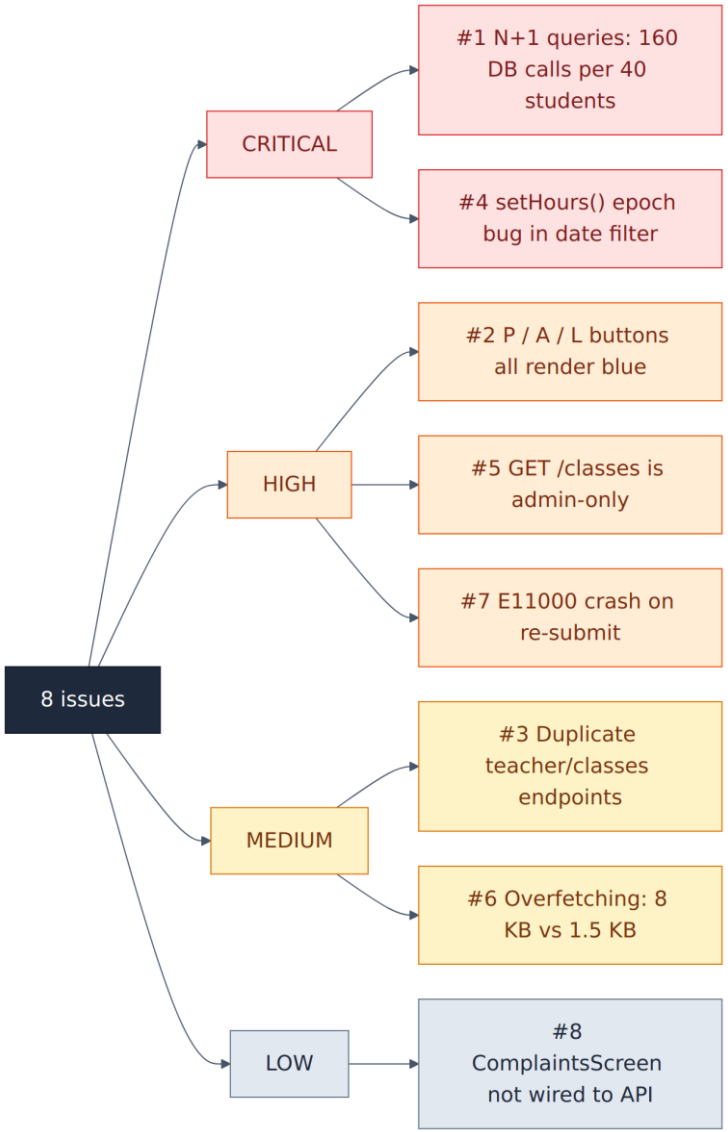


The Attendance tab is the core of the assessment: date bar, class card, live summary, searchable student list, and one Submit button.

5.2 Database entity relationships



6. Issues, bugs and fixes



6.1 Duplicate API routes: before and after

